5.1.4:

Solution:

(c) i.) The characteristic polynomial for A is

$$\lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$$

ii.) For $\lambda = 1$:

$$A - I = \begin{pmatrix} i-1 & 1 \\ 2 & -1-i \end{pmatrix} \Rightarrow \begin{cases} (i-1)x_1 + x_2 & = 0 \\ 2x_1 + (-1-i)x_2 & = 0 \end{cases} \Rightarrow x_2 = (1-i)x_1 \Rightarrow X_1 = t\begin{pmatrix} 1 \\ 1-i \end{pmatrix}$$

For $\lambda = -1$:

$$A + I = \begin{pmatrix} i+1 & 1 \\ 2 & 1-i \end{pmatrix} \Rightarrow \begin{cases} (i+1)x_1 + x_2 & = 0 \\ 2x_1 + (1-i)x_2 & = 0 \end{cases} \Rightarrow x_2 = -(i+1)x_1 \Rightarrow X_2 = t\begin{pmatrix} 1 \\ -1-i \end{pmatrix}$$

iii.) Since X_1 and X_2 are linearly independent,

$$\left\{ \begin{pmatrix} 1 \\ 1-i \end{pmatrix}, \begin{pmatrix} 1 \\ -1-i \end{pmatrix} \right\}$$

is a basis for $\mathbb{C}^2$.

iv.) It follows that

$$Q = \begin{pmatrix} 1 & 1 \\ 1-i & -1-i \end{pmatrix} \quad \text{and} \quad D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

so Q^{-1} exists since $\det(Q) = -2 \neq 0$ and $Q^{-1}AQ = D$.

(d) i.) The characteristic polynomial is

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 0 & -1 \\ 4 & 1-\lambda & -4 \\ 2 & 0 & -\lambda - 1 \end{vmatrix} = (1-\lambda)\Big((2-\lambda)(-\lambda-1) + 2\Big) = 0$$

$$\Rightarrow (1-\lambda)(\lambda^2 - \lambda) = 0$$
$$\Rightarrow -\lambda(\lambda-1)^2 = 0$$
$$\Rightarrow \lambda = 0, 1 \,(\text{mul. } 2)$$

Hence A has eigenvalues $\lambda = 0, 1$.

ii.) For $\lambda = 0$:

$$A - 0I = \begin{pmatrix} 2 & 0 & -1 \\ 4 & 1 & -4 \\ 2 & 0 & -1 \end{pmatrix} \xrightarrow[r_2 \to r_2 - 2r_1]{r_3 \to r_3 - r_1} \begin{pmatrix} 2 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} x_3 & = t \\ x_2 - 2x_3 & = 0 \\ 2x_1 - x_3 & = 0 \end{cases} \Rightarrow X_1 = t\begin{pmatrix} 1/2 \\ 2 \\ 1 \end{pmatrix}$$

For $\lambda = 1$:

$$A - I = \begin{pmatrix} 1 & 0 & -1 \\ 4 & 0 & -4 \\ 2 & 0 & -2 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} x_3 & = t \\ x_2 & = s \\ x_1 - x_3 & = 0 \end{cases} \Rightarrow X_2 = t\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + s\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

iii.) Since eigenvectors corresponding to distinct eigenvalues are linearly independent (and the two vectors spanning the eigenspace for $\lambda = 1$ are independent),

$$\left\{ \begin{pmatrix} 1/2 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

is a basis for $\mathbb{R}^3$.

iv.) Hence

$$Q = \begin{pmatrix} 1/2 & 1 & 0 \\ 2 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \quad \text{and} \quad D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and $\det(Q) = 0.5 \neq 0$ so Q^{-1} exists and $D = Q^{-1}AQ$.

5.1.5*f*:

Solution: Let γ be the standard ordered basis for $P_3(\mathbb{R})$ so

$$[T]_\gamma = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 3 & 4 & 8 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then the characteristic polynomial is

$$\det([T]_\gamma - \lambda I) = \begin{vmatrix} 1-\lambda & 0 & 0 & 0 \\ 1 & 3-\lambda & 4 & 8 \\ 0 & 0 & 1-\lambda & 0 \\ 0 & 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda) \begin{vmatrix} 3-\lambda & 4 & 8 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^3(3-\lambda) = 0$$

so T has eigenvalues 1 (mul. 3) and 3. For $\lambda = 1$:

$$[T]_\gamma - I = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 2 & 4 & 8 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} x_4 & = t \\ x_3 & = r \\ x_1 + 2x_2 + 4x_3 + 8x_4 & = 0 \Rightarrow x_2 = -4t - 2r - \frac{1}{2}s \\ x_1 & = s \end{cases}$$

Hence

$$X = s \begin{pmatrix} 1 \\ -\frac{1}{2} \\ 0 \\ 0 \end{pmatrix} + r \begin{pmatrix} 0 \\ -2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ -4 \\ 0 \\ 1 \end{pmatrix}.$$

Now, for $\lambda = 3$:

$$[T]_\gamma - 3I = \begin{pmatrix} -2 & 0 & 0 & 0 \\ 1 & 0 & 4 & 8 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 4 & 8 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{cases} x_4 & = 0 \\ x_3 & = 0 \\ x_2 & = t \\ x_1 & = 0 \end{cases}$$

Hence,

$$X = t \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

Thus, the ordered basis is $\beta = \left\{1 - \frac{x}{2}, x^2 - 2x, x^3 - 4x, x\right\}$ and

$$[T]_\beta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

is clearly diagonal.

5.1.9:

Solution:

(a) *Proof.* ($\Rightarrow$) By way of the contrapositive assume that 0 is an eigenvalue of T. It follows by definition that $\det(A - 0I) = \det(A) = 0$ where $A = [T]_\beta$ for any ordered basis β. However, it then follows that A is not invertible by the Corollary to Thm 4.7 and so T cannot be invertible by Thm 2.18.
($\Leftarrow$) Conversely, suppose that 0 is not an eigenvalue of T and, for the sake of contradiction, that T is not invertible. Then $A = [T]_\beta$ is not invertible for any ordered basis β by Thm 2.18, so $\det(A) = 0$ by the Corollary to Thm 4.7. Hence $\det(A - 0I) = 0$, so 0 is a root of the characteristic polynomial of T and thus an eigenvalue of T, a contradiction. Therefore T is invertible. □

(b) *Proof.* ($\Rightarrow$) Let $T : V \to V$ be an invertible linear operator such that $\lambda^{-1} \in \mathbb{F}$ is an eigenvalue of T^{-1} (here $\lambda \neq 0$ by part (a)). By definition it follows that there exists some nonzero $w \in V$ such that

$$T^{-1}(w) = \lambda^{-1}w \Rightarrow T(T^{-1}(w)) = T(\lambda^{-1}w) \Rightarrow w = \lambda^{-1}T(w) \Rightarrow \lambda w = T(w)$$

and hence λ must be an eigenvalue of T.
($\Leftarrow$) Conversely we may follow an almost identical argument assuming λ is an eigenvalue for T with eigenvector $w \neq 0$ (note $\lambda \neq 0$ by part (a)):

$$T(w) = \lambda w \Rightarrow T^{-1}(T(w)) = T^{-1}(\lambda w) \Rightarrow w = \lambda T^{-1}(w) \Rightarrow \lambda^{-1}w = T^{-1}(w)$$

and hence λ^{-1} is an eigenvalue for T^{-1}. □

5.1.10:

Solution:

Proof. Let $M \in \mathbb{M}_{n\times n}(\mathbb{F})$ be any upper triangular matrix. Then,

$$\det(M - \lambda I) = \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} - \lambda & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} - \lambda & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} - \lambda \end{vmatrix} = (a_{11} - \lambda)(a_{22} - \lambda)\cdots(a_{nn} - \lambda)$$

using the fact that the determinant of any upper triangular matrix is the product of its diagonal entries (Exercise 4.2.23). Hence, the eigenvalues of M are exactly its diagonal entries. □

5.2.2:

Solution:

(d) We start by finding the characteristic polynomial

$$\det(A-\lambda I)=\begin{vmatrix}7-\lambda & -4 & 0\\ 8 & -5-\lambda & 0\\ 6 & -6 & 3-\lambda\end{vmatrix}=(3-\lambda)\Big((7-\lambda)(-5-\lambda)+32\Big)$$
$$=(3-\lambda)(\lambda^2-2\lambda-3)$$
$$=(3-\lambda)(\lambda+1)(\lambda-3)=0$$
$$\Rightarrow \lambda=3\,(\text{mul. }2),-1$$

So clearly the characteristic polynomial splits over $\mathbb{R}$. For $\lambda_1=3$:

$$A-3I=\begin{pmatrix}4&-4&0\\8&-8&0\\6&-6&0\end{pmatrix}\to\begin{pmatrix}1&-1&0\\0&0&0\\0&0&0\end{pmatrix}\Rightarrow\begin{cases}x_3&=t\\x_2&=r\\x_1&=x_2\end{cases}\Rightarrow X=r\begin{pmatrix}1\\1\\0\end{pmatrix}+t\begin{pmatrix}0\\0\\1\end{pmatrix}$$

thus

$$E_{\lambda_1}=\left\{r\begin{pmatrix}1\\1\\0\end{pmatrix}+t\begin{pmatrix}0\\0\\1\end{pmatrix}\mid r,t\in\mathbb{R}\right\}.$$

For $\lambda_2=-1$:

$$A+I=\begin{pmatrix}8&-4&0\\8&-4&0\\6&-6&4\end{pmatrix}\to\begin{pmatrix}8&-4&0\\0&0&0\\6&-6&4\end{pmatrix}\Rightarrow\begin{cases}6x_1-6x_2+4x_3&=0\\x_2&=2t\\8x_1-4x_2&=0\end{cases}\Rightarrow X=t\begin{pmatrix}1\\2\\3/2\end{pmatrix}$$

thus

$$E_{\lambda_2}=\left\{t\begin{pmatrix}1\\2\\3/2\end{pmatrix}\mid t\in\mathbb{R}\right\}.$$

Since $\dim(E_{\lambda_1})=2=$ multiplicity of $\lambda_1=3$ and $\dim(E_{\lambda_2})=1=$ multiplicity of $\lambda_2=-1$, A is diagonalizable. So,

$$Q=\begin{pmatrix}1&0&1\\1&0&2\\0&1&3/2\end{pmatrix}\quad\text{and}\quad D=\begin{pmatrix}3&0&0\\0&3&0\\0&0&-1\end{pmatrix}$$

(f) Since A is upper triangular we know that its eigenvalues are its diagonal entries: $1\,(\text{mul. }2)$ and 3. Since $1,3\in\mathbb{R}$ the characteristic polynomial must split over $\mathbb{R}$. For $\lambda_1=1$:

$$A-I=\begin{pmatrix}0&1&0\\0&0&2\\0&0&2\end{pmatrix}\to\begin{pmatrix}0&1&0\\0&0&1\\0&0&0\end{pmatrix}\Rightarrow\begin{cases}x_3&=0\\x_2&=0\\x_1&=t\end{pmatrix}\Rightarrow X=t\begin{pmatrix}1\\0\\0\end{pmatrix}$$

so

$$E_{\lambda_1}=\left\{t\begin{pmatrix}1\\0\\0\end{pmatrix}\mid t\in\mathbb{R}\right\}.$$

Since $\dim(E_{\lambda_1})=1\neq 2=$ multiplicity of $\lambda_1=1$, A is not diagonalizable.

5.2.3*d*:

Solution: Let α be the standard ordered basis of $P_2(\mathbb{R})$. Then

$$[T]_\alpha = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = A \Rightarrow \det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix} = (1-\lambda)\Big((1-\lambda)^2 - 1\Big)$$
$$= (1-\lambda)(1 - 2\lambda + \lambda^2 - 1)$$
$$= (1-\lambda)(\lambda^2 - 2\lambda) = 0$$
$$\Rightarrow \lambda = 0, 1, 2$$

Since $0, 1, 2 \in \mathbb{R}$ the characteristic polynomial splits over $\mathbb{R}$. Now, for $\lambda_1 = 0$:

$$A \to \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} x_3 &= t \\ x_2 &= -t \\ x_1 &= 0 \end{cases} \Rightarrow E_{\lambda_1} = \left\{ t \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \mid t \in \mathbb{R} \right\}.$$

For $\lambda_2 = 1$:

$$A - I = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{cases} x_3 &= t \\ x_2 &= t \\ x_1 &= -t \end{cases} \Rightarrow E_{\lambda_2} = \left\{ t \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \mid t \in \mathbb{R} \right\}.$$

For $\lambda_3 = 2$:

$$A - 2I = \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$
$$\Rightarrow \begin{cases} x_3 &= t \\ x_2 &= t \\ x_1 &= 0 \end{cases} \Rightarrow E_{\lambda_3} = \left\{ t \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \mid t \in \mathbb{R} \right\}.$$

Hence since the dimension of each eigenspace equals the multiplicity of its respective eigenvalue, T is diagonalizable. Thus, for

$$\beta = \left\{ x^2 - x, x^2 + x - 1, x + x^2 \right\}$$

$[T]_\beta = \text{diag}(0, 1, 2)$ is diagonal.

5.2.13:

Solution:

(a) *Proof.* Consider an invertible linear operator T and let λ be one eigenvalue of T so λ^{-1} is an eigenvalue of T^{-1}. Consider the eigenspace of T corresponding to λ, E_λ, and the eigenspace of T^{-1} corresponding to λ^{-1}, $E'_{\lambda^{-1}}$ (note that $\lambda \neq 0$ by Exercise 5.1.9). Now, let $v \in E_\lambda$ be any vector in the eigenspace of λ, so that $T(v) = \lambda v$. Then,

$$T(v) = \lambda v \Rightarrow T^{-1}(T(v)) = T^{-1}(\lambda v) \Rightarrow v = \lambda T^{-1}(v) \Rightarrow \lambda^{-1} v = T^{-1}(v)$$

so $v \in E'_{\lambda^{-1}}$ and since v was arbitrary, $E_\lambda \subseteq E'_{\lambda^{-1}}$. An almost identical argument can be made by flipping λ and λ^{-1}, T and T^{-1}, and E_λ and $E'_{\lambda^{-1}}$ to show $E'_{\lambda^{-1}} \subseteq E_\lambda$. Hence, $E_\lambda = E'_{\lambda^{-1}}$ as was to be shown. □

(b) *Proof.* It follows from (a) that every eigenvector of T is an eigenvector of T^{-1}, with eigenvalue replaced by its reciprocal. Suppose that T is diagonalizable. Then there is a basis β of V consisting of eigenvectors of T, and by (a) each vector of β is also an eigenvector of T^{-1}. Hence β is a basis of V consisting of eigenvectors of T^{-1}, so T^{-1} is diagonalizable. □

5.2.21:

Solution:

Proof. ($\Rightarrow$) Suppose that V is the direct sum of $W_1, \dots, W_k$ and let β_i be a basis for W_i. We claim that the list β obtained by concatenating $\beta_1, \dots, \beta_k$ is a basis for V. It spans V since $\sum W_i = V$. For independence, suppose a linear combination of the vectors in β equals 0 and let $w_i \in W_i$ be the part of that combination coming from β_i, so $w_1 + \cdots + w_k = 0$. Then for each j,

$$w_j = -\sum_{i \neq j} w_i \in W_j \cap \sum_{i \neq j} W_i = \{0\},$$

so every $w_j = 0$, and since each β_i is linearly independent, every coefficient is 0. Hence β is a basis for V. In particular β has no repeated vectors, so

$$\dim(V) = |\beta| = \sum_{i=1}^{k} |\beta_i| = \sum_{i=1}^{k} \dim(W_i).$$

($\Leftarrow$) Now suppose that $\dim(V) = \sum_{i=1}^{k} \dim(W_i)$ and let β be the list obtained by concatenating $\beta_1, \dots, \beta_k$ where each β_i is a basis for W_i. Since $\sum W_i = V$, clearly β spans V, and since β has exactly $\sum_{i=1}^{k} \dim(W_i) = \dim(V)$ vectors, β must be a basis for V. In particular β is linearly independent. Now, consider $x \in W_j \cap \sum_{i \neq j} W_i$ so

$$x = a_1 v_1 + \cdots + a_m v_m \quad \text{and} \quad x = b_1 u_1 + \cdots + b_p u_p \Rightarrow 0 = (a_1 v_1 + \cdots + a_m v_m) - (b_1 u_1 + \cdots + b_p u_p)$$

for $v_1, \dots, v_m \in \beta_j$ and $u_1, \dots, u_p \in \bigcup_{i \neq j} \beta_i$ and scalars $a_1, \dots, a_m, b_1, \dots, b_p \in \mathbb{F}$. The vectors $v_1, \dots, v_m, u_1, \dots, u_p$ are distinct vectors of β, so since β is linearly independent all of the scalars must be 0. In particular $a_1 = \cdots = a_m = 0$ and so $x = 0$. Thus $W_j \cap \sum_{i \neq j} W_i = \{0\}$ for every j, and so V is the direct sum of $W_1, \dots, W_k$, as was to be shown. □

5.2.22:

Solution:

Proof. Define $W_i = \operatorname{span}(\beta_i)$ so $W_i \subseteq V$. Since $\beta_i \cap \beta_j = \varnothing$ for $i \neq j$, then $|\beta| = \sum |\beta_i|$ but $|\beta| = \dim(V)$, and $|\beta_i| = \dim(W_i)$ since β_i is linearly independent (being a subset of the basis β) and spans W_i, so

$$\dim(V) = |\beta| = \sum |\beta_i| = \sum \dim(W_i).$$

As well, clearly

$$\sum_{i=1}^{k} W_i = V$$

since β_i partition β so by exercise (5.2.21),

$$V = W_1 \oplus \cdots \oplus W_k = \operatorname{span}(\beta_1) \oplus \cdots \oplus \operatorname{span}(\beta_k)$$

as was to be shown. □